SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA[illegible]

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL 2)

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 25

Code of Conduct

I Kaviya · G (192524281) (Name / Reg. No.)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Kaviya · G

Assessment Tasks
Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

Total Marks:

Assessment Tool 1

Short Answer Test

1) Define cloud computing and give four characteristics

Cloud computing is the delivery of computing services like storage, servers, software and databases through the Internet

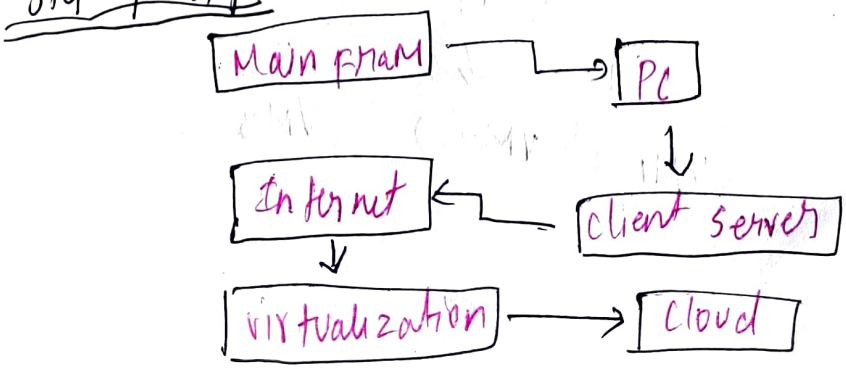
Characteristics:

1) on demand: resources are available whenever needed
2) internet access: services can be accessed from anywhere
3) resource sharing: resource are shared among many users
4) scalability: resources can be increased or decreased easily

---

2) Evolution of cloud computing

step by step



Virtualization	cloud computing
Technology	service model
Creates virtual machine	Provides service through Internet
Uses hypervisor	user virtualization
Eg: VMware	Eg: AWS

H/ Role of data centers

Internet

A data center is a place containing servers, storage, networking, power, cooling systems

cloud services

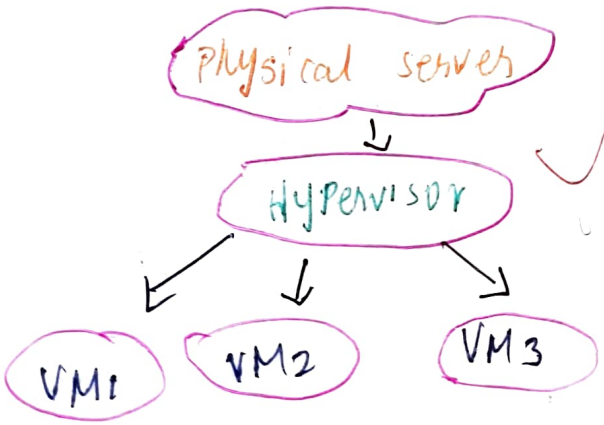
* Stores user data
* Runs cloud application
* Provides computing Resources
* Maintains security and backup
* Supports high availability

Eg: google drive store files in google's data centers

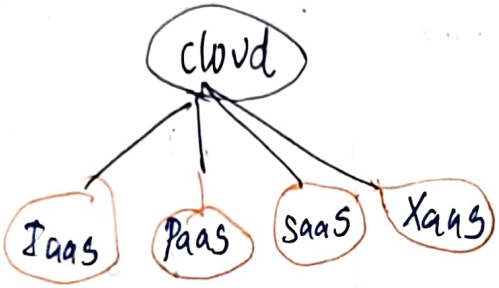
* Main frames provided shared computing
* PCs made computing available to individuals
* Client server allowed sharing through networks
* Internet enabled access from anywhere
* virtualization allowed one server to Run Many virtual Machine
* web servis provided software through the Internet
* these technologies together created Modern cloud Platforms

3) Server virtualization and cloud computing

server virtualization divides one physical server into multiple virtual Machine (VMs)



3) Iaas, Paas, Saas, xaas
↓ ↓ ↓ ↓
Infrastructure Platform Software servic
cloud servic Model



Model	Meaning	Example
Iaas	Provide infrastructure like servis and storage	AWS EC2
Paas	provides a platform a develop application	Google app engine
Saas	provide cloud to use software	Gmail
Xaas	Provides any IT servic through Internet	Dropbox

2/10/26